DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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Subject: Python Programming		
Assignment No.: 2		
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Lab Assignment: 02

Aim:
To write a Python program to enter a number and display its hexadecimal equivalent, octal equivalent, and square root.

Objective:
- To understand and use Python's built-in functions for number system conversion.
- To convert a given decimal number into its hexadecimal and octal equivalents.
- To apply the math module for performing mathematical operations like finding the square root.
- To gain practical experience in accepting user input and displaying formatted output in Python.

Theory:
Python provides built-in functions to convert numbers into different number systems.
- hex() converts a decimal number to hexadecimal.
- oct() converts a decimal number to octal.

The math module provides mathematical functions.
- math.sqrt() is used to calculate the square root of a number.

The round() function is used to limit the number of decimal places.

Algorithm:
1. Start the program.
2. Import the math module.
3. Input an integer number from the user.
4. Convert the number into hexadecimal using hex().
5. Convert the number into octal using oct().
6. Find the square root using math.sqrt().
7. Display the hexadecimal value, octal value, and square root.
8. Stop the program.

Pseudo Code:
```
START
IMPORT math
READ number
```

hex_value ← hex(number)
oct_value ← oct(number)
sqrt_value ← sqrt(number)
DISPLAY hex_value
DISPLAY oct_value
DISPLAY sqrt_value (rounded)
END

Code:

```
import math;
num=int(input("Enter a number: "));
print("Hexadeciaml equivalent: ",hex(num));
print("Octal equivalent: ",oct(num));
num1=math.sqrt(num);
print("Square Root: ",round(num1,2));
```

Output:

```
Enter a number:  7
Hexadeciaml equivalent:  0x7
Octal equivalent:  0o7
Square Root:  2.65
```

Conclusion:
The program successfully accepts a number from the user and displays its hexadecimal, octal, and square root values using Python's built-in functions and the math module. This demonstrates number system conversion and mathematical operations in Python.